

M52: Final Composite Dependent Variable

PCA on Raw Switch Counts (Time, Topic-LDA, Typing)

No standardization (z-score) applied

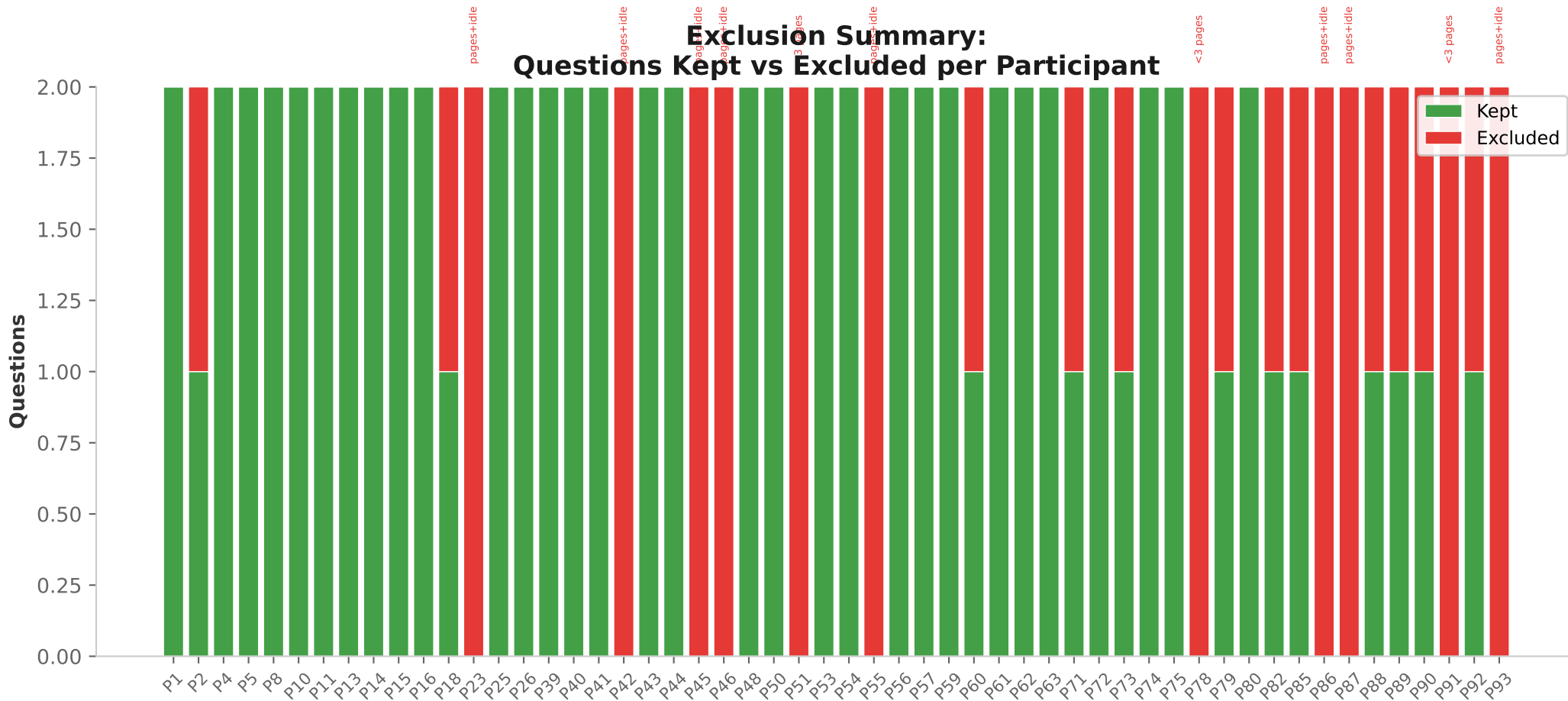
Exclusion Criteria Applied:

1. Questions with fewer than 3 page visits: 32 excluded
2. Questions with idle time $\geq 50\%$ after last meaningful event: 15 excluded
3. Participants > 3 SD from mean on any DV: 0 excluded

Result: 74 / 108 questions retained

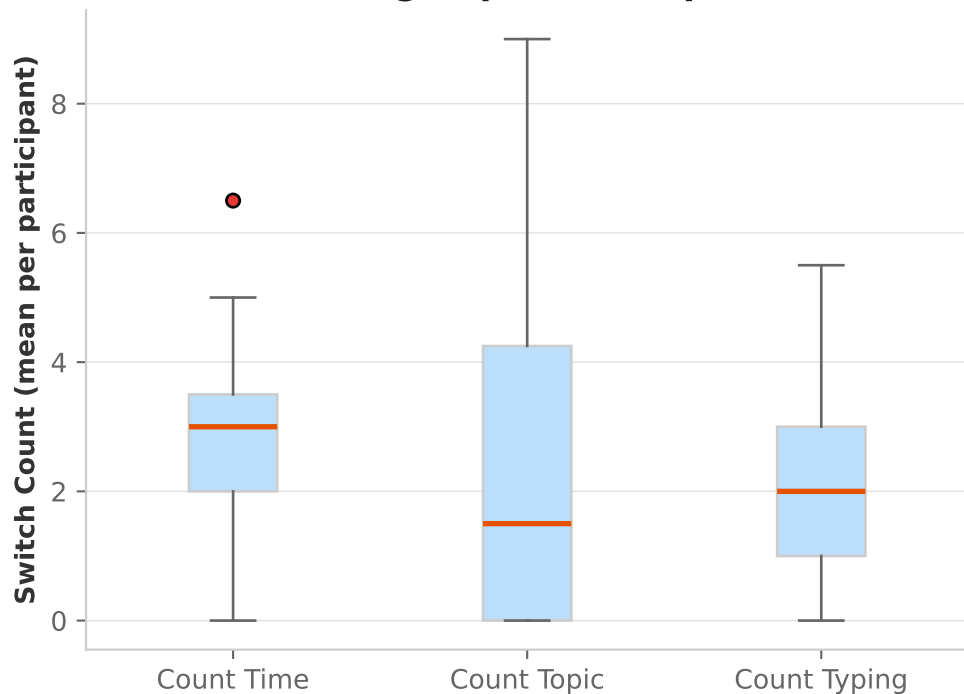
Final sample: N = 43 participants

Exclusion Summary: Questions Kept vs Excluded per Participant

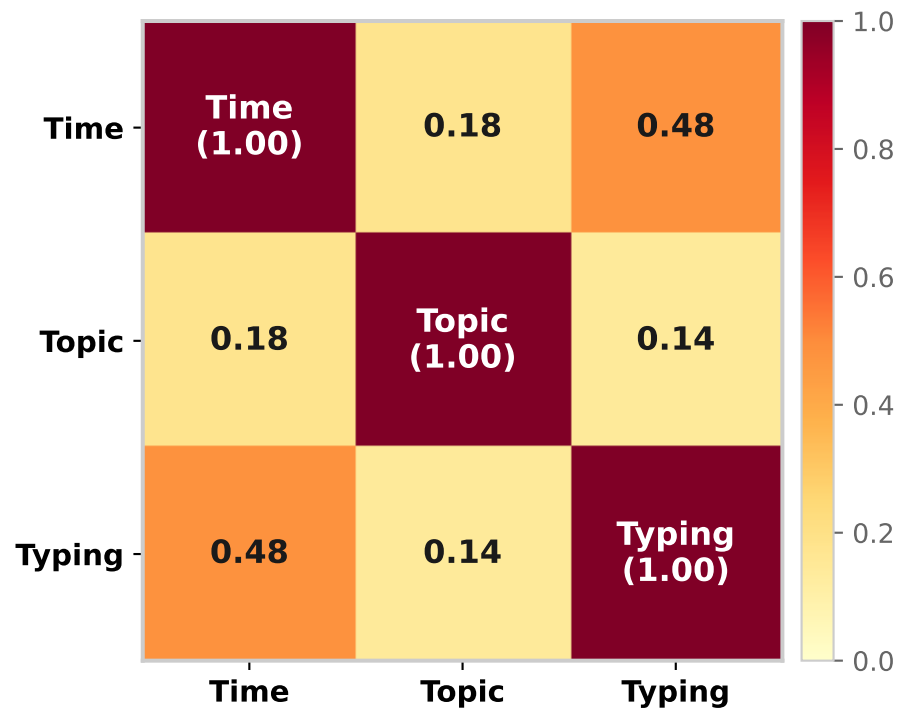


Descriptive Statistics: Raw Switch Counts

Distribution of Raw Switch Counts (Averaged per Participant)

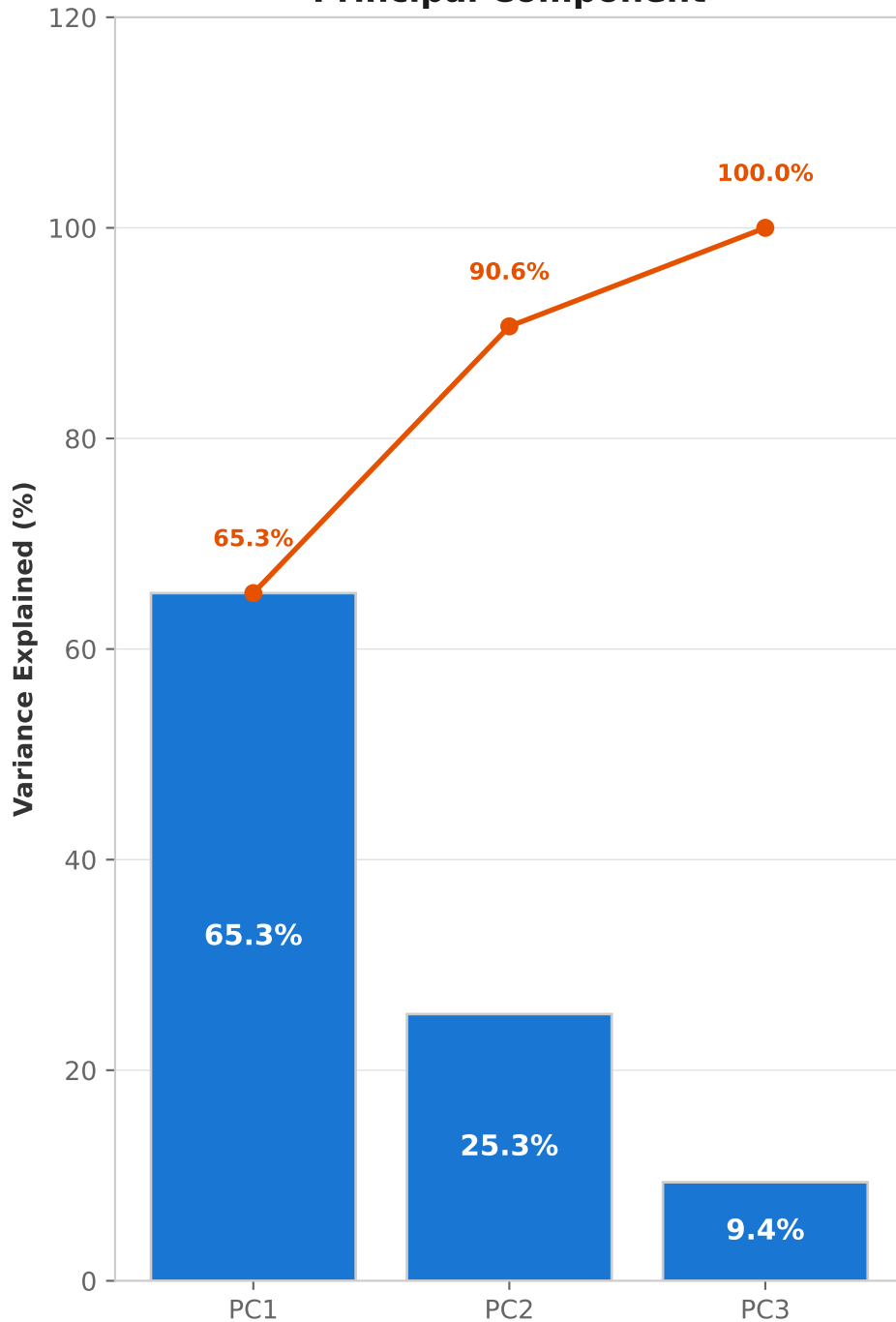


Correlation Between Switch Count Variables



PCA Results: Dimensionality Reduction on Raw Switch Counts

Variance Explained by Each Principal Component



PCA Loadings (variable weights per component):

PC1 (65.3% variance):

Count Topic	+0.981	+++++
Count Time	+0.150	+++
Count Typing	+0.126	++

PC2 (25.3% variance):

Count Time	+0.697	+++++
Count Typing	+0.690	+++++
Count Topic	-0.195	---

PC3 (9.4% variance):

Count Typing	+0.713	+++++
Count Time	-0.701	-----
Count Topic	+0.016	

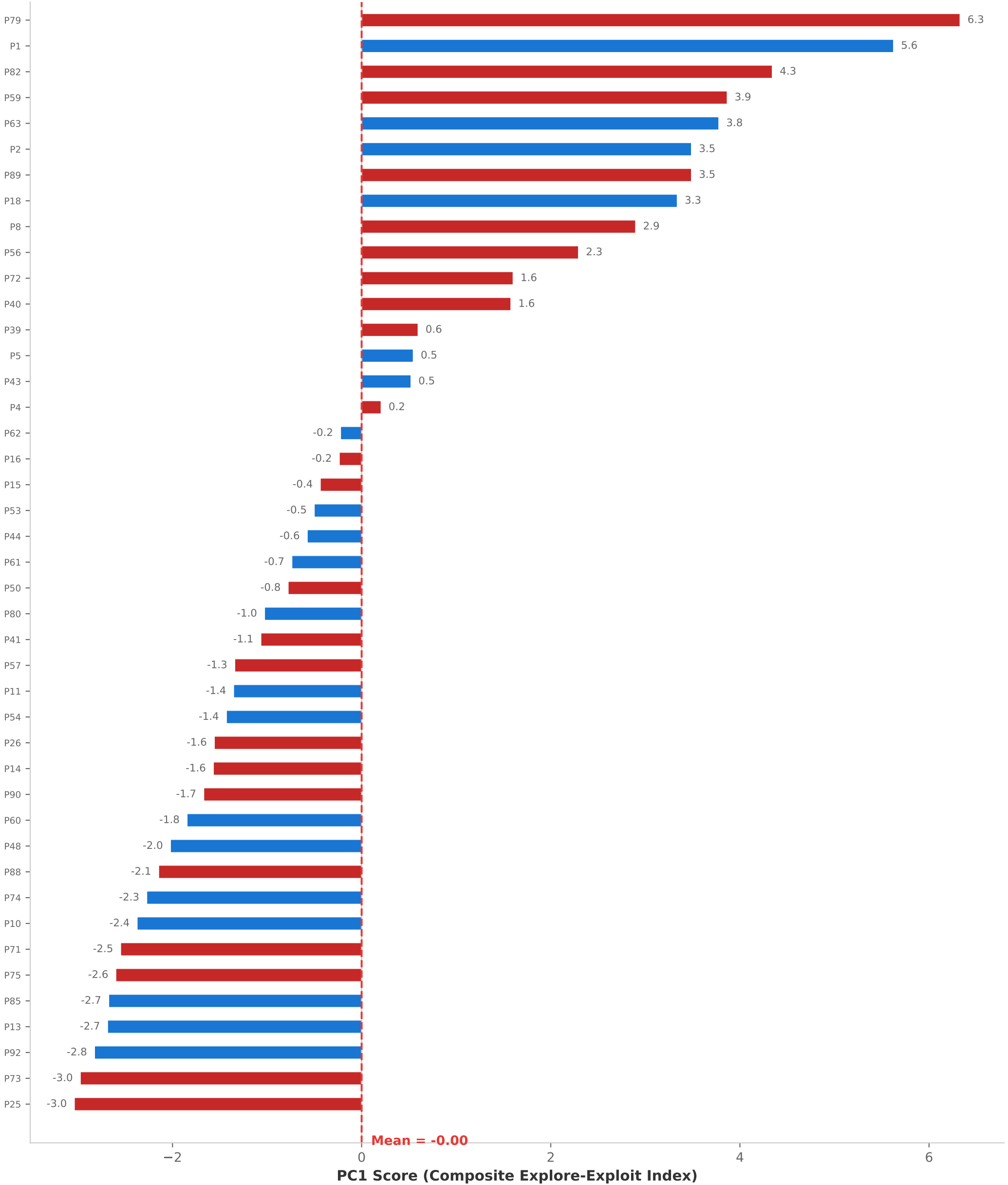
Interpretation:

PC1: Primarily reflects topic switching
(loading = +0.981)
Higher score = more topic transitions

PC2: Primarily reflects time switching
(loading = +0.697)
Higher score = more time transitions

PC3: Primarily reflects typing switching
(loading = +0.713)
Higher score = more typing transitions

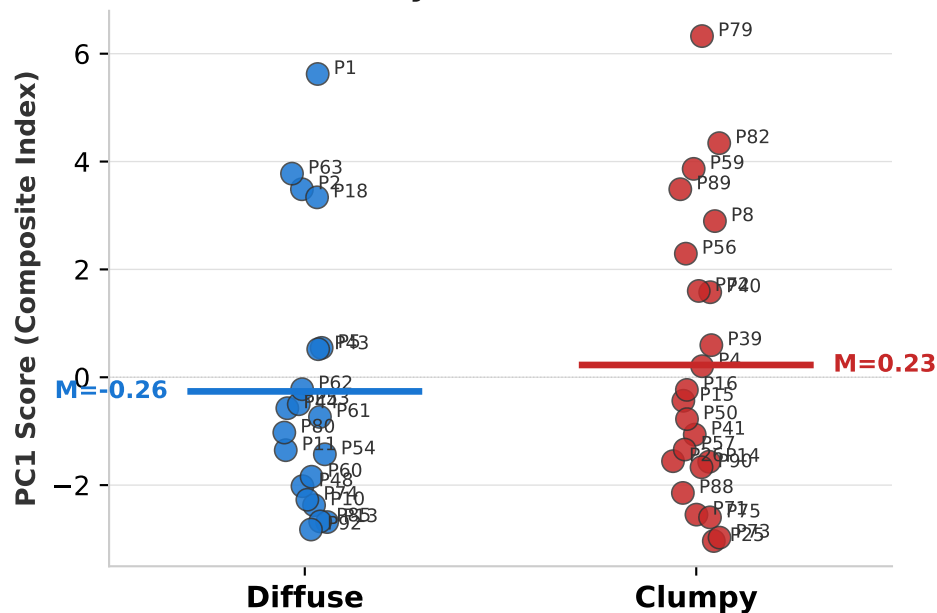
Individual PC1 Scores: Composite Explore-Exploit Index



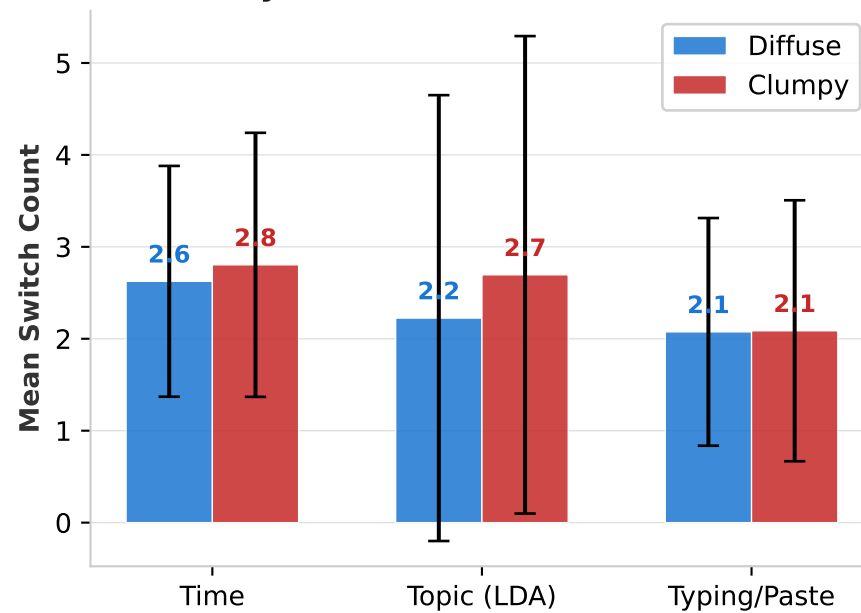
Condition Comparison: Clumpy vs Diffuse

N = 43 (Diffuse: 20, Clumpy: 23)

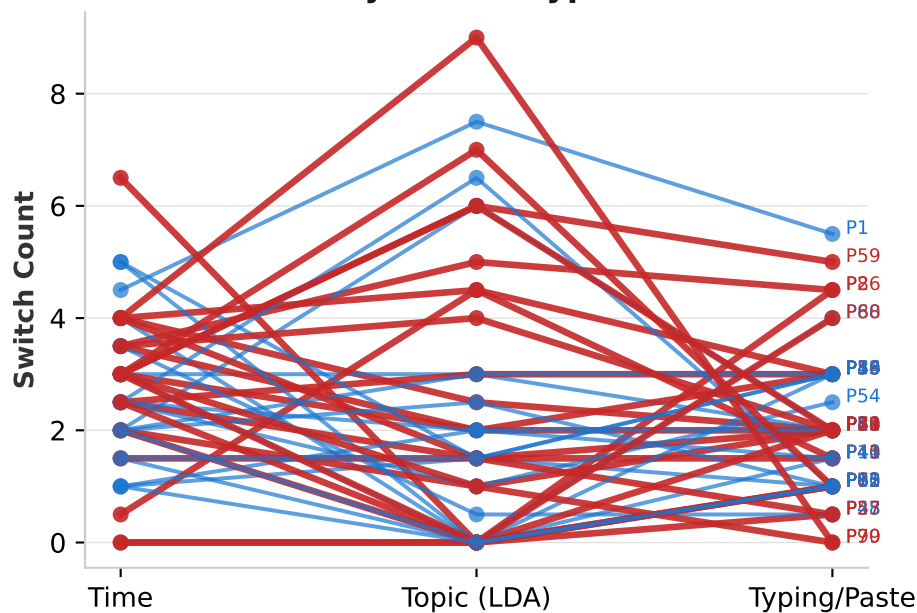
PC1: Composite Explore-Exploit
by Condition



Raw Switch Counts
by Condition (Mean +/- SD)



Individual Profiles
by Switch Type



Conditions

- Diffuse (n=20)
- Clumpy (n=23)

Condition Comparison: Descriptive Statistics & t-Tests

N = 43 (Diffuse: 20, Clumpy: 23)

DESCRIPTIVE STATISTICS

Variable	Diffuse (n=20)	Clumpy (n=23)
Time	M = 2.62, SD = 1.26	M = 2.80, SD = 1.44
Topic (LDA)	M = 2.23, SD = 2.43	M = 2.70, SD = 2.60
Typing/Paste	M = 2.08, SD = 1.24	M = 2.09, SD = 1.42
PC1 (composite)	M = -0.26, SD = 2.46	M = 0.23, SD = 2.62

INDEPENDENT-SAMPLES t-TESTS (Welch)

Variable	t	df	p	Cohen d	Sig.
Time	-0.44	41.0	0.6644	+0.13	n.s.
Topic (LDA)	-0.61	40.8	0.5425	+0.19	n.s.
Typing/Paste	-0.03	41.0	0.9766	+0.01	n.s.
PC1 (composite)	-0.63	40.7	0.5311	+0.19	n.s.

* p < .05 ** p < .01 *** p < .001

INTERPRETATION

PC1 difference (Clumpy - Diffuse): +0.49
Clumpy condition shows more explore-exploit switches
Effect size (Cohen's d): +0.19
Result is NOT statistically significant (p = 0.5311)
Low statistical power due to small sample (N = 43)

Largest sub-variable effect: Topic (LDA)
d = +0.19, p = 0.5425

Note: Typing metric uses answer_snapshot bursts + paste events.